AWS PROJECT REPORT

VPC PEERING

Connecting private resources across two Amazon VPCs

Student Name	Harsha Vardhan Arle
Internship Batch	AWS INTERNSHIP BATCH 1
Project	AWS VPC Peering
AWS Region	Asia Pacific (Mumbai), ap-south-1
Submission Date	30 October 2026

Prepared as the final documentation for the AWS internship practical project.

1. Project Overview

The objective of this project was to create two separate Amazon Virtual Private Clouds (VPCs), connect them using VPC Peering, and verify private communication between EC2 instances. A public bastion host in VPC A provides an SSH entry point, while the application instances remain in private subnets without public IPv4 addresses.

Expected outcome: App-A in VPC A and App-B in VPC B can communicate over private IP addresses through the peering connection. A laptop can also SSH to the private instances through the bastion host using SSH ProxyJump.

2. Architecture Diagram

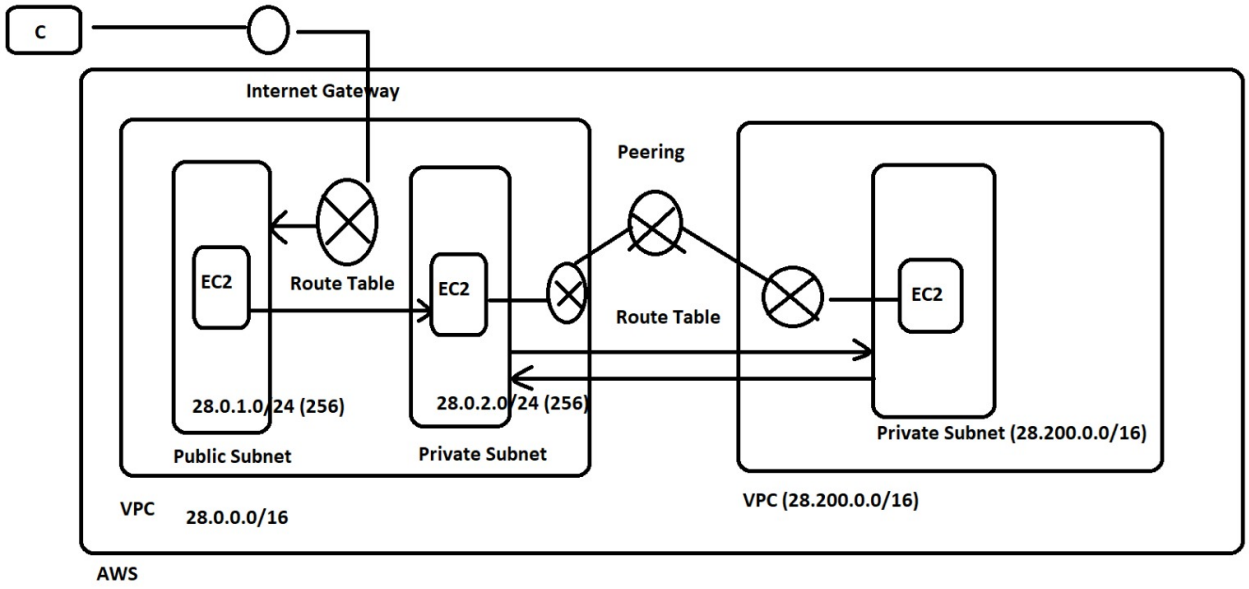


Figure 1. Original architecture diagram supplied from the internship explanation class.

3. AWS Environment and Resource Inventory

All resources were created in the Mumbai region (ap-south-1). The following inventory records the names and addressing plan used in the implementation.

Resource	Name	Configuration
VPC A	ewb-vpc-a	10.10.0.0/16
VPC B	ewb-vpc-b	10.20.0.0/16
Public subnet	ewb-a-public	10.10.1.0/24, VPC A
Private subnet A	ewb-a-private	10.10.2.0/24, VPC A
Private subnet B	ewb-b-private	10.20.1.0/24, VPC B
Internet Gateway	ewb-igw-a	Attached to VPC A
Public route table	ewb-rt-a-public	Associated with ewb-a-public
Private route table A	ewb-rt-a-private	Associated with ewb-a-private
Private route table B	ewb-rt-b-private	Associated with ewb-b-private
Peering connection	ewb-a-b-peer	Active
Key pair	ewb-lab	RSA; private PEM stored locally

4. Routing and VPC Peering Configuration

The two VPCs use non-overlapping IPv4 CIDR ranges. The peering connection ewb-a-b-peer was accepted and reached Active status. Routes were then added to both private route tables, and to the public route table in VPC A so the bastion host can route traffic to App-B through the peering connection.

Route table	Destination	Target / purpose
ewb-rt-a-public	10.10.0.0/16	local
ewb-rt-a-public	0.0.0.0/0	Internet Gateway ewb-igw-a
ewb-rt-a-public	10.20.0.0/16	VPC peering connection ewb-a-b-peer
ewb-rt-a-private	10.10.0.0/16	local
ewb-rt-a-private	10.20.0.0/16	VPC peering connection ewb-a-b-peer
ewb-rt-b-private	10.20.0.0/16	local
ewb-rt-b-private	10.10.0.0/16	VPC peering connection ewb-a-b-peer

The VPC peering connection is not a gateway and does not provide transitive routing. VPC B does not use VPC A's Internet Gateway for internet access.

5. EC2 Instance Configuration

Instance	Private IP	Subnet	Public IPv4
ewb-bastion	10.10.1.10	ewb-a-public	Assigned
ewb-app-a	10.10.2.10	ewb-a-private	None
ewb-app-b	10.20.1.10	ewb-b-private	None

All three instances used Amazon Linux 2023 and instance type t3.micro. The bastion host was placed in the public subnet and had a public IPv4 address. App-A and App-B were placed in their respective private subnets and had no public IPv4 addresses.

6. Security Group Configuration

The security groups were configured to permit the required SSH and ICMP traffic. The bastion allowed SSH from the student's current public IP address using a /32 source. The application instances allowed SSH from the bastion private IP (10.10.1.10/32) and ICMP Echo Request from the opposite application instance's private IP.

Security group	Inbound access documented
ewb-sg-bastion	SSH TCP 22 from student's public IP /32
ewb-sg-app-a	SSH TCP 22 from 10.10.1.10/32; ICMP Echo Request from 10.20.1.10/32
ewb-sg-app-b	SSH TCP 22 from 10.10.1.10/32; ICMP Echo Request from 10.10.2.10/32

Outbound rules were configured to allow all traffic to 0.0.0.0/0 for the bastion and both application security groups. Additional SSH access from the respective EC2 Instance Connect Endpoint security groups was temporarily added for troubleshooting access. These temporary rules and endpoints should be removed during cleanup if they are no longer required.

7. Connectivity Validation and Results

The following tests were completed and verified during the practical work:

Test	Command / action	Observed result
App-A to App-B	ping -c 4 10.20.1.10	4 packets received; 0% packet loss
App-B to App-A	ping -c 4 10.10.2.10	4 packets received; 0% packet loss
Laptop to App-A	ssh ewb-app-a	Successful SSH login through bastion using ProxyJump
Laptop to App-B	ssh ewb-app-b	Successful SSH login through bastion using ProxyJump
Controlled failure	Temporarily remove App-B inbound ICMP Echo Request rule from 10.10.2.10/32; ping App-B from App-A	4 packets transmitted; 100% packet loss
Recovery	Restore the ICMP rule and repeat ping	4 packets received; 0% packet loss

The controlled failure demonstrated that the ICMP security group rule was required for ping traffic. Restoring that rule restored connectivity. SSH access remained configured separately from ICMP access.

8. Key Learnings

• VPC peering connects two VPCs using private IP routing, provided the CIDR ranges do not overlap.
• An Active peering connection alone is not sufficient. Routes must be configured on both sides.
• Security groups are stateful and must allow the required inbound traffic. Outbound rules must also permit the intended traffic.
• A private EC2 instance can be reached from a laptop through a bastion host using SSH ProxyJump, without assigning the private instance a public IP address.
• VPC peering is non-transitive and does not make VPC B use VPC A's Internet Gateway.

9. Screenshot Evidence Appendix

The following screenshot placeholders are provided so the student's original console and terminal captures can be inserted later. The screenshots were not embedded in this generated PDF because they were not uploaded with the report request. Replace each placeholder with the matching screenshot file before final submission.

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR
–		Available	–	–	Off	
ewb-vpc-a	vpc-0fc0a87441906a5e3	Available	–	–	Off	10.10.0.0/16
ewb-vpc-b	vpc-0807505c5fa68e523	Available	–	–	Off	10.20.0.0/16

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
ewb-a-private	subnet-01781aae2fb56118a	Available	vpc-0fc0a87441906a5e3 | ewb-...	Off	10.10.2.0/24
ewb-a-public	subnet-07976772a8ace0523	Available	vpc-0fc0a87441906a5e3 | ewb-...	Off	10.10.1.0/24
–		Available		Off	
ewb-b-private	subnet-05369ff92f00c19e9	Available	vpc-0807505c5fa68e523 | ewb-...	Off	10.20.1.0/24

Name	Internet gateway ID	State	VPC ID	Owner
–		Attached		
ewb-igw-a	igw-0f4b1a2bb28428cf8	Attached	vpc-0fc0a87441906a5e3 | ewb-vpc-a	

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
ewb-rt-b-private	rtb-045c3b39101efb8f0	subnet-05369ff92f00c19...	–	No	vpc-0807505c5fa68e523 | ewb-...
–		–	–		
–		–	–		
ewb-rt-a-private	rtb-0d4326e645db0c516	subnet-01781aae2fb561...	–	No	vpc-0fc0a87441906a5e3 | ewb-...
ewb-rt-a-public	rtb-00edd67536ad96290	subnet-07976772a8ace0...	–	No	vpc-0fc0a87441906a5e3 | ewb-...

rtb-00edd67536ad96290 / ewb-rt-a-public

Routes (3)

Both | Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0f4b1a2bb28428cf8	Active	No	Create Route
10.10.0.0/16	local	Active	No	Create Route Table
10.20.0.0/16	pcx-04b76475ced6fc363	Active	No	Create Route

rtb-0d4326e645db0c516 / ewb-rt-a-private

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (2)

Both | Edit routes

Filter routes

1

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	local	Active	No	Create Route Table
10.20.0.0/16	pcx-04b76475ced6fc363	Active	No	Create Route

rtb-045c3b39101efb8f0 / ewb-rt-b-private

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (2)

Both | Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.10.0.0/16	pcx-04b76475ced6fc363	Active	No	Create Route
10.20.0.0/16	local	Active	No	Create Route Table

Name	Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs	Accepter CIDRs
ewb-a-b-peer	pcx-04b76475ced6fc363	Active	vpc-0fc0a87441906a5e3 / ewb...	vpc-0807505c5fa68e523 / ewb...	10.10.0.0/16	10.20.0.0/16

Name	Security group ID	Security group name	VPC ID
ewb-sg-app-a	sg-058b234a3fb5c047a	ewb-sg-app-a	vpc-0fc0a87441906a5e3
ewb-sg-bastion	sg-0d12be8c0f373493b	ewb-sg-bastion	vpc-0fc0a87441906a5e3
–		default	
ewb-sg-app-b	sg-0ffea76d375190074	ewb-sg-app-b	vpc-0807505c5fa68e523

sg-0d12be8c0f373493b - ewb-sg-bastion

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (1)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
–	sgr-05fe4102becdd0d53	IPv4	SSH	TCP	22	/32

sg-058b234a3fb5c047a - ewb-sg-app-a

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (3)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
–	sgr-0895145affbabd0d8	IPv4	SSH	TCP	22	10.10.1.10/32
–	sgr-050d1de569faddbd3	–	SSH	TCP	22	sg-0a256c81e66c021ff ...
–	sgr-0849479774fa894e4	IPv4	Custom ICMP - IPv4	Echo Request	N/A	10.20.1.10/32

sg-0ffea76d375190074 - ewb-sg-app-b

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Inbound rules (3)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
–	sgr-03c299d82c1623dc5	IPv4	All ICMP - IPv4	ICMP	All	10.10.2.10/32
–	sgr-05d2bdef8ea0ec8f5	–	SSH	TCP	22	sg-0c18222fa9499237d...
–	sgr-0479a93499e00b810	IPv4	SSH	TCP	22	10.10.1.10/32

i-018b24ea2ae95865b (ewb-bastion)

Instance ID
i-018b24ea2ae95865b

IPv6 address
–

Hostname type
IP name: ip-10-10-1-10.ap-south-1.compute.internal

Instance type
t3.micro

Public IPv4 address
[redacted] | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-10-1-10.ap-south-1.compute.internal

Elastic IP addresses
–

Private IPv4 addresses
10.10.1.10

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
[redacted] [Public IP]

i-095c9463d7d96639a (ewb-app-a)

Instance ID
i-095c9463d7d96639a

IPv6 address
–

Hostname type
IP name: ip-10-10-2-10.ap-south-1.compute.internal

Instance type
t3.micro

Public IPv4 address
–

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-10-2-10.ap-south-1.compute.internal

Elastic IP addresses
–

Private IPv4 addresses
10.10.2.10

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
–

i-06c708b6ed0e91c9b (ewb-app-b)

Instance ID
i-06c708b6ed0e91c9b

IPv6 address
–

Hostname type
IP name: ip-10-20-1-10.ap-south-1.compute.internal

Instance type
t3.micro

Public IPv4 address
–

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-20-1-10.ap-south-1.compute.internal

Elastic IP addresses
–

Private IPv4 addresses
10.20.1.10

Public DNS
–

Answer private resource DNS name
–

Auto-assigned IP address
–

```
Harsha@Harsha:~$ ssh ewb-app-a
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Oct    04:56:51 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.561 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.558 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.555 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.615 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3149ms
rtt min/avg/max/mdev = 0.555/0.572/0.615/0.024 ms
[ec2-user@ip-10-10-2-10 ~]$
```

```
Harsha@Harsha:~$ ssh ewb-app-b
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Oct    05:01:53 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.539 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.543 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.529 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.701 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3123ms
rtt min/avg/max/mdev = 0.529/0.578/0.701/0.071 ms
[ec2-user@ip-10-20-1-10 ~]$
```

```
Harsha@Harsha:~$ ssh ewb-app-a
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
Last login: Sat Oct 11 06:20:20 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$
```

```
Harsha@Harsha:~$ ssh ewb-app-b
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
Last login: Sat Oct 11 06:23:12 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$
```

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3127ms

[ec2-user@ip-10-10-2-10 ~]$
```

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.546 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.549 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.548 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.557 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3148ms
rtt min/avg/max/mdev = 0.546/0.550/0.557/0.004 ms
[ec2-user@ip-10-10-2-10 ~]$
```

sg-0d12be8c0f373493b - ewb-sg-bastion

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Outbound rules (1)

Manage tags | Edit outbound rules

Search

Security group rule ID	IP version	Type	Protocol	Port range	Destination
sgr-07bc0566fce36c841	IPv4	All traffic	All	All	0.0.0.0/0

sg-058b234a3fb5c047a - ewb-sg-app-a

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Outbound rules (1)

Manage tags | Edit outbound rules

Search

Security group rule ID	IP version	Type	Protocol	Port range	Destination
sgr-053e55e1b63cc76ee	IPv4	All traffic	All	All	0.0.0.0/0

sg-0ffea76d375190074 - ewb-sg-app-b

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - *new* | Tags

Outbound rules (1)

Manage tags | Edit outbound rules

Search

Security group rule ID	IP version	Type	Protocol	Port range	Destination
sgr-0b15b22571f47bc7f	IPv4	All traffic	All	All	0.0.0.0/0

rtb-00edd67536ad96290 / ewb-rt-a-public

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Explicit subnet associations (1)

Edit subnet associations

Find subnet association

1

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ewb-a-public	subnet-07976772a8ace0523	10.10.1.0/24	–

rtb-0d4326e645db0c516 / ewb-rt-a-private

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Explicit subnet associations (1)

Edit subnet associations

Find subnet association

1

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ewb-a-private	subnet-01781aae2fb56118a	10.10.2.0/24	–

rtb-045c3b39101efb8f0 / ewb-rt-b-private

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Explicit subnet associations (1)

Edit subnet associations

Find subnet association

1

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ewb-b-private	subnet-05369ff92f00c19e9	10.20.1.0/24	–

10. Cleanup Checklist

After the report and evidence have been saved, remove AWS resources to avoid ongoing charges. Do not begin cleanup until you have finished any required demonstration and saved all evidence.

Recommended cleanup order:

1. Terminate the three EC2 instances: ewb-bastion, ewb-app-a, and ewb-app-b.
2. Check and remove any remaining EBS volumes or snapshots that are not required.
3. Delete both EC2 Instance Connect Endpoints created for troubleshooting, if no longer needed.
4. Remove temporary EIC Endpoint security group inbound rules from the application security groups. Do not remove the intended bastion SSH or ICMP rules until instances are terminated.
5. Delete the VPC peering connection and remove peering routes from all three route tables.
6. Delete custom security groups after dependent network interfaces are gone. Do not delete a shared default security group.
7. Disassociate and delete the subnets and custom route tables as required.
8. Detach and delete ewb-igw-a.
9. Delete ewb-vpc-a and ewb-vpc-b.
10. Verify that no project resources remain. Billing information may take time to update.

11. Conclusion

The project successfully demonstrated private communication between two Amazon VPCs using VPC Peering. App-A and App-B exchanged ICMP Echo Requests in both directions, and both private instances were reachable from the laptop through the bastion host using SSH ProxyJump. The controlled failure and recovery exercise demonstrated the effect of the relevant security group rule on network connectivity.

Security note: The private key file (ewb-lab.pem), passwords, access keys, and other credentials must not be included in this report or screenshots.

End of Report